



PICK AND PLACE ARM – USER MANUAL

Introduction

This 3D-printed robotic arm is designed as an educational robotic system for learning about servo motors, joystick control, microcontrollers, and robotic arm mechanisms. The arm uses four servo motors to control the **base, shoulder, elbow, and gripper**. The 3D-printed enclosure provides the mechanical structure for mounting and protecting the electronic components while allowing the robotic arm to perform controlled movements.

Main Features

3D-printed robotic arm structure	Elbow movement
Four servo motor-based movements	Gripper open/close operation
Dual joystick control	Automatic return-to-center for base, shoulder, and elbow
Base rotation	Smooth servo movement
Shoulder movement	Joystick dead-zone control

Components Required

No.	Component	Quantity
1	Arduino Nano	1
2	Servo Motor	4
3	Joystick Module	2
4	3D-Printed Robotic Arm Enclosure/Parts	1 Set
5	Jumper Wires	As required
7	USB Cable	1
8	PCB Board	1

Connections

Servo Motor Connections

Servo	Function	Arduino Pin
Servo 1	Base	D2
Servo 2	Shoulder	D3
Servo 3	Elbow	D5
Servo 4	Gripper	D6

Joystick Connections

Joystick	Axis	Function	Arduino Pin
Joystick 1	X-axis	Base	A0
Joystick 1	Y-axis	Shoulder	A1
Joystick 2	X-axis	Gripper Open/Close	A2
Joystick 2	Y-axis	Elbow	A3

Assembly Steps

Step 1: Connect the Base Servo

Connect the **base servo motor** to the **base part** as shown in the picture. Align the servo properly and secure it using the provided screws.



Step 2: Connect the Shoulder Servo

Connect the **shoulder servo motor** to the **shoulder part**. Position it correctly and securely fasten the servo using the screws.



Step 3: Connect the Elbow and Gripper

Connect the **elbow servo** and **gripper assembly** to the **3D-printed parts** as shown in the image. Make sure all parts are properly aligned and secure them using the provided screws.



Step 4: Connect the Servo Wires

Connect the **red wires** of all the servo motors to the **VCC** terminal on the PCB. Connect the **brown wires** of all the servo motors to the **GND** terminal on the PCB. Use **jumper wires** to make the required connections and ensure that all connections are secure.

Step 5: Connect the Servo Signal Wires

Connect the **servo signal wires (yellow wires)** to the corresponding **servo pins** on the PCB according to the **connection table provided above**. Ensure that each servo is connected to the correct pin and that all connections are secure.

Step 6: Place and Connect the Joystick

Place the **joystick module** in the designated position as shown in the picture. Place the **enclosure over the PCB board** and ensure that it is properly aligned and secured. Connect the joystick to the **PCB** according to the **connection table provided above**. Ensure that all wires are connected to the correct pins and are securely fitted.



Control Instructions

Joystick 1

Move **LEFT/RIGHT** → Rotate the robotic arm base.

Move **UP/DOWN** → Move the shoulder.

Release the joystick → Base/shoulder gradually return to the **90° original position**.

Joystick 2

Move **UP/DOWN** → Move the elbow.

Release the joystick → Elbow returns to the **90° original position**.

Move **LEFT** → Open the gripper.

Move **RIGHT** → Close the gripper.

How to Operate

1. Place the robotic arm on a stable and flat surface.
2. Check all servo motor and joystick connections.
3. Connect the Arduino to the computer using a USB cable.
4. Upload the robotic arm program to the Arduino.
5. Provide a suitable external power supply for the servo motors.
6. Wait for the arm to move to its initial position.
7. Use **Joystick 1** to control the base and shoulder.
8. Use **Joystick 2** to control the elbow and gripper.
9. Move the joysticks gently to achieve smooth movement.
10. After completing the operation, return the arm to a safe position and switch OFF the power.

Safety Precautions

- ❖ Keep fingers away from moving joints and the gripper.
- ❖ Do not force the robotic arm beyond its mechanical limits.
- ❖ Do not manually rotate the servo arms while power is ON.
- ❖ Use a suitable power supply for the servo motors.
- ❖ Ensure the 3D-printed parts are securely assembled.
- ❖ Check that screws and servo mounts are properly tightened.
- ❖ Do not place heavy objects beyond the arm's capacity.
- ❖ Keep wires away from moving joints.
- ❖ Switch OFF the power before changing any wiring.
- ❖ Do not operate the arm on an unstable or slippery surface.
- ❖ Stop the system immediately if a servo makes unusual noise or becomes excessively hot.

Learning Outcomes

Students will learn the basic concepts of robotic arm mechanisms, servo motor control, joystick interfacing, Arduino programming, and 3D-printed mechanical design. They will understand how joystick inputs are converted into servo movements and how multiple components can be integrated to operate a Pick and Place Arm. Students will also develop practical skills in circuit connections, assembly, programming, and troubleshooting.

Conclusion

The 3D-printed robotic arm provides a practical platform for understanding robotics, electronics, programming, and mechanical design. The dual-joystick control allows students to interact directly with the robotic arm and understand how input commands are converted into precise servo movements.

The project can also be further upgraded with **Bluetooth/Wi-Fi control, object detection, automatic pick-and-place operations, sensors, and AI-based control**.